DIETCOST IN R

Reference Manual

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1 INTRODUCTION

DIETCOST is a free open-source software developed to evaluate the cost and environmental impact of an ideal diet, through Monte Carlo simulations. First introduced by the pioneer study of Vandevijvere *et al*. (2018), the method was initially made available in the Python programming language.

Aiming to ease the access to the software, this new version was constructed as a R Shiny application turned into na executable file, prescinding any programming knowledge. Also, tending to the needs of users who demanded more customable approaches, a companion R library, named *DIETCOST* (2025), was developed and uploaded to CRAN, the oficial repository of the R language.

The remainer of this document is organized as it follows: a quick theoretical background of the method is provided in Chapter 2, while the technical info of this new version is presented in Chapter 3. Chapter 4 consists of a general overview of the app, and each tab is discussed in detail in Chapters 5-7.

2 THEORETICAL BACKGROUND

This software uses a Monte Carlo framework to assess the nutritional, cost and environmental parameters of a diet. The process is quickly described in this section. Following the assembling of a commonly available foods database, a random meal plan is created. Its nutritional value is confronted to the nutrient and food group serves targets of a given individual and diet, i.e. a 37-year old woman in a healthy diet. Also taken in consideration are the serves of linked foods – foods whose consumption is evaluated together, with a lower bracket whose serves are at maximum equal to a higher bracket.

The difference between the recently created plan and the set of constraints is then calculated. If its value is not zero, a random non-null nutrient/food group is selected, and a random food which affects said parameter has its intake randomly altered, closing a single iteration. At the beginning of the next one, the nutrients of the meal plan are re-calculated and the process begins anew, to the limit of the number of interations imputed by the user. A diet is formed when all the constraints are observed, resulting in a net value of zero to the diferences matrix.

3 TECHNICAL ASPECTS

This application was developed in the *R* language (v.4.4.1), using the *RStudio* IDE (v. 2023.09.0 – Build 463). The main library used was *shiny* (v. 1.9.1), a web application framework for R. The source code is available at its dedicated GitHub repository: https://github.com/hbracarense/DIETCOST-App.

4 GENERAL OVERVIEW

A dashboard approach with a navegation bar at the topside was a selected, with individual tabs that compartimentalize the steps needed to treat the datasets before the simulation. Besides the **Introduction** one, there are three tbas, which will be explored in their own-dedicated chapters: **Foods**, **Constraints** and **Simulation**.

5 FOODS TAB

The Foods tab refers to the commonly available foods database. The user can choose between submitting their own file to the software, following the **standard table model** adopted by the program, or assembling through a graphic interface. There aren't any limits to this selection, although the validity of the results is compromised by significantly small databases. Also, if the user chooses to submit an Excel spreadsheet in this tab, they will be forced to adopt the same procedure in the next tab.

6 CONSTRAINTS TAB

The access to this tab is only permitted after the user complets the **Foods** form. If he chose to upload their data in the previous section, the same will be required here; once again, the standard model is made available. A sanity check is throughly performed, with the main requirements being equivalence between the foods in both tabs and the same nutrient columns and food groups in the main database and in the constraints.

In the other hand, if the graphic interface was chosen, the user can adopt a pre-saved set consisting of a profile (45-years man, 37-years woman, 12-years boy or 8-years girl) and a diet (current, healthy or EAT-Lancet) or set their own constraints through the interface.

The mandatory constraints are foods and food groups requirements and nutrient targets. Regard the latter, there are 17 possible constraints: basic nutrientes (energy, fat, saturated fat, sugars, protein, fibre and sodium), percentagy of energy derived from these nutrientes, red meat intake and percentage of energy acquired through the intake of special groups (alcohol, discretionary foods and takeaway). The minimum set of choices is one, although the same caveat made above is still valid.

Finally, the linked foods constraint is optional: the user can choose one pair, two pairs or neither. Again, two forms of selection are possible: the pre-sets (**milk/cereal** and **bread/cream**), if any two foods, one from either bracket, are in the dataset oro ne defined by the user, also in the maximum of two pairs.

7 SIMULATION TAB

Regardless of the path taken in the two precedent tabs, the interface here is the same. The random meal created in the first access of this tab (once again permitted only when the two previous forms are completed) is displayed as a table. A folder to receive the diets formed, exported as .csv files, will be created; the user must choose the main directory where it will be held. Two numeric parameters are required: the number of iterations to be performed (max 1 billion) and the minimum serve size difference. Finally, the user can select which of the parameters (price, carbon footprint, water footprint and/or ecological footprint), if any of them is available in the data, will be evaluated. Once the simulation begins, it can be stopped by pressing the **Stop** button. After it finishes, the enitre app can be reset and the reports concerning the run, also with the basic statistics of the variables, if any diet is formed, can be exported as na Excel workbook.

8 CONTACT INFO

Any queries, demands or bug reports can be submitted either at the GitHub repositor or directly to h@bracarense.com.

REFERENCES

Bracarense, H., Marquezine, T. and Claro, R. DIETCOST: Calculate the Cost and Environmental Impact of a Ideal Diet. R package version 1.0.0.0 (2025). https://CRAN.R-project.org/package=DIETCOST

Vandevijvere, S., Young, N., Mackay, S. et al. Modelling the cost differential between healthy and current diets: the New Zealand case study. Int J Behav Nutr Phys Act 15, 16 (2018). https://doi.org/10.1186/s12966-018-0648-6